

Answer all questions

QUESTION 1

Considering the **digit** dataset from **Sklearn** library, demonstrate with the elbow technique that 10 is the best number of clusters when the **KMEANS** algorithm is applied (10 marks)

QUESTION 2

Import **CIFAR10** dataset from **TENSORFLOW.KERAS** library

1. Build a convolutional Neural Network made up of three (03) stacks, Each stack is made up of:
 - a. one convolutional layer,
 - b. one RELU layer,
 - c. one Pulling layer.

The set of stacks should be followed by a fully connected layer with a hidden layer. with the aid of summary command display the model architecture (10 marks)

2. Train the model for 5 epochs and Plot the confusion Matrix for the testing set (10 marks)

Question 3

With the use of any Deep Learning library (**OpenCV, YoLO, Teachable machine, Media Pipe**)

1. Describe your solution and give the steps to go through in a flowchart (5marks)
2. **deploy a model to to identify features or persons from either image or video stream** (15marks)